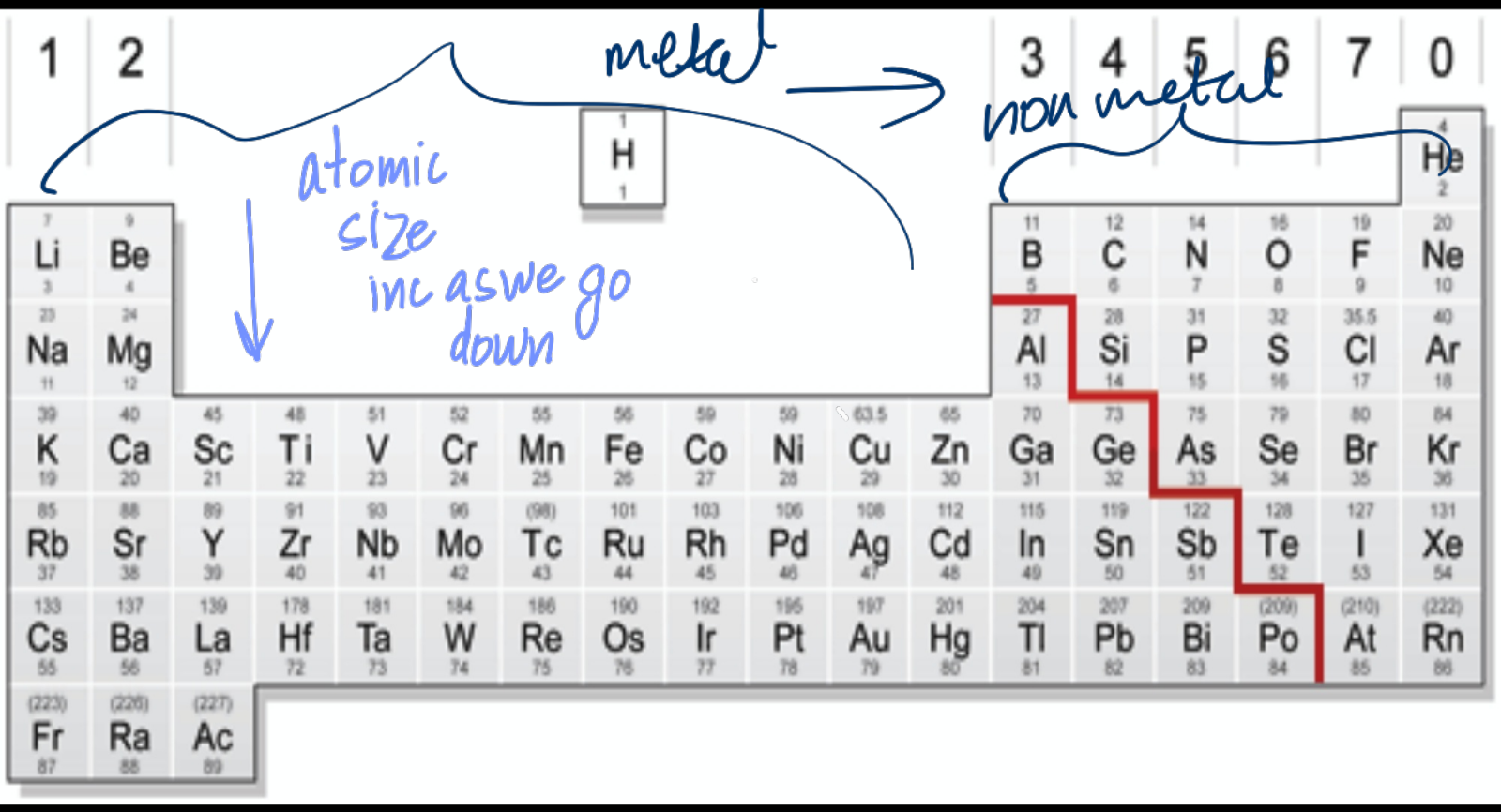
Understanding periodic Table

— Notes by Ishani

Basic properties



Group 1 → Alkali metals -

(3) Lithium (Li) → 2, 1
(11) Sodium (Na) → 2, 8, 1
(19) Pottasium (K) → 2, 8, 8, 1

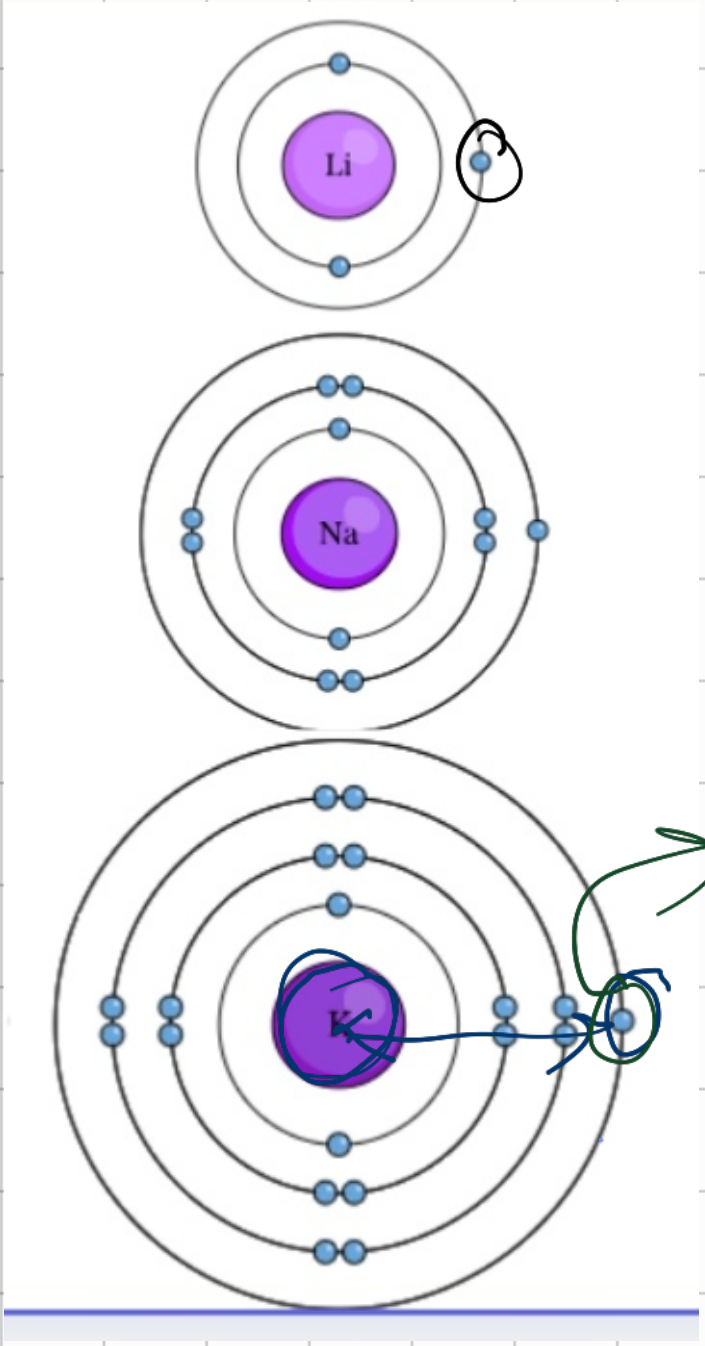
- 1 electron in outermost shell : Group 1
- These metals are soft and can be cut with knife
- **Relatively low mp and low density**
- **Highly reactive**
- Stored in oil to prevent them react with reacting with oxygen and water vapour

Three main trends -

(K) Boiling Point, Melting point decreases down the group

(K) Density increases as we go down the group

(K) Reactivity increases as we go down the group



Size increases as we go down

Weaker attraction btw the nucleus and outer shell electron

∴ less energy is required to remove electron from element further down group

⇒ Reactivity increases as we go down



Li



Na



K

Alkali metal	Gas production	Heat produced by reaction	Production of alkaline solution	Production of flame
lithium	fizzes slowly, a few bubbles	moves slowly on the surface, does not melt	metal disappears slowly, solution turns purple	no flame
sodium	fizzes strongly, many bubbles	melts into a ball on surface and moves about	metal disappears quickly, solution turns purple	no flame unless held in place
potassium	fizzes violently, very many bubbles	melts and moves very quickly on surface	disappears very quickly, solution turns purple	hydrogen ignites, a lilac flame produced

Table 13.3: Observations on the reactions of the alkali metals with water (with universal indicator added).

Group 1 elements react with cold water to form Hydrogen and alkaline solution.

The production of alkaline solution can be prooved by adding few drops of universal indicator to water

The solution turns purple

Eg: $2Na + H_2O \longrightarrow 2NaOH + H_2$

Sodium + water → Sodium hydroxide + hydrogen

Paper 3

Questions are applicable for both core and extended candidates

1 The chemical elements are arranged in the Periodic Table in groups and periods.

(a) (i) Describe how the metallic character of the elements changes from left to right across a period.

decreases

..... [1]

(ii) The elements in Group I are known as the alkali metals.

Describe **two** trends in the properties of the elements, going down Group I.

1 density inc

2 reactivity inc

[2]

2 This question is about metals.

(a) Iron is a transition element. Potassium is an element in Group I of the Periodic Table.

State **two** differences in the physical properties of iron compared to potassium.

1 .....

2 .....

[2]

3 This question is about metals and metal compounds.

(a) Table 5.1 shows some properties of some Group I metals.

Table 5.1

metal	melting point in °C	boiling point in °C	observations on reaction with water	solubility of metal hydroxide in g/dm^3 at room temperature
sodium	98	883	bubbles form rapidly but no flame	
potassium	63	760		1130
rubidium		686	explodes	1980
caesium	29	669	explodes	3860

Use the information in Table 5.1 to predict: btw 63 and 29 (excluding 63 & 29)

later

(i) the melting point of rubidium ..... btw 62 and 30 (including 62 & 30) [1]

(ii) ~~the solubility of sodium hydroxide at room temperature~~ ..... [1]

(iii) the observations when potassium reacts with water

..... bubbles very rapidly AND flames → also ✓

..... [1]

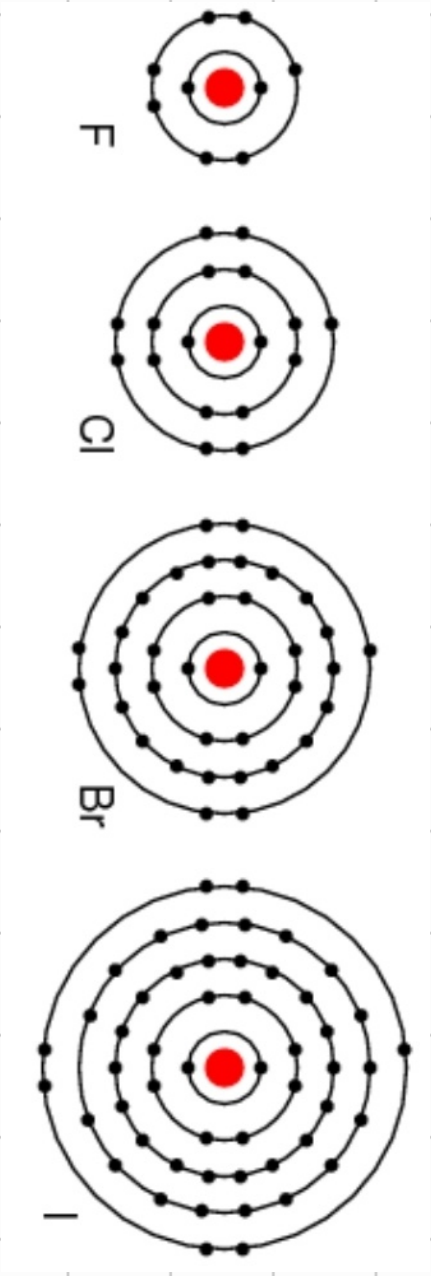
(iv) the physical state of caesium at 20 °C. Give a reason for your answer.

physical state ..... solid .....

reason ..... 20°C is below M.P ∴ it's a solid .....

..... [2]

★ Group 7 → THE HALOGENS



↓ Size increases as we go down

Attraction btw outer electrons and nucleus decreases
So it is harder to gain electron

(1) So, Reactivity decreases ★

→ imp ★★

3 Trends –

(I) Density increases as we go down the group
(II) Reactivity decreases as we go down
(III) BP and MP increases as we go down ★

⊛ Appearance - (Room temp)

very imp

F_2	Yellow gas
Cl_2	Green gas
Br_2	Reddish brown liquid
I_2	Grey solid
At_2	Black solid



Only liquid non metal at room temp

Intensity of colour of elements increases from pale to dark as we go down

Iodine

Grey solid / Reddish brown sol in water / Sublimes to form water vapours







The more reactive non metal displaces less reactive non metal from its salt

Eg Clorine + Pottasium bromide → Bromine + Pottasium Cloride

$Cl_2 + 2KBr \rightarrow Br_2 + 2KCl$

$2KI \rightarrow I_2 + 2KCl$

orange

The halogen displacement reactions shown in the lower part of Table 13.5 demonstrate the order of reactivity of the three major halogens. For example, if you add chlorine to a solution of potassium bromide, the chlorine displaces bromine (Figure 13.10). Chlorine is more reactive than bromine, so it replaces bromine and potassium chloride is formed. Potassium bromide solution is colourless. It turns orange when chlorine is bubbled through it:

→ $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$ gas

colourless — orange

Chlorine will also displace iodine from potassium iodide:

$Cl_2 + 2KI \rightarrow 2KCl + I_2$

colourless — yellow–brown

KEY WORDS

halogen displacement reactions: reactions in which a more reactive halogen displaces a less reactive halogen from a solution of its salt

1 The chemical elements are arranged in the Periodic Table in groups and periods.

(b) Chlorine, bromine and iodine are in Group VII of the Periodic Table.

(i) Aqueous chlorine reacts with aqueous sodium bromide to produce aqueous bromine and aqueous sodium chloride.

$Cl_2 + 2\ NaBr \rightarrow Br_2 + 2NaCl$

Complete the symbol equation for this reaction.

$Cl_2 + __.NaBr \rightarrow ___ + 2NaCl$ [2]

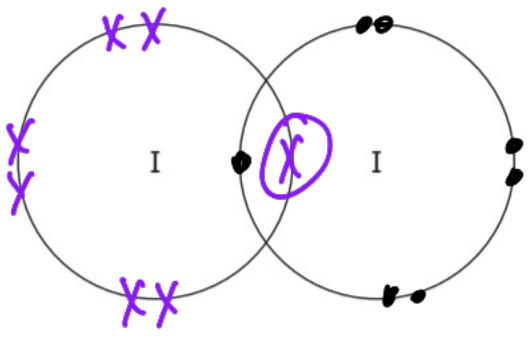
(ii) Suggest why aqueous iodine does **not** react with aqueous sodium bromide.

.....

iodine less reactive than Bromine ..... [1]

(iii) Complete the dot-and-cross diagram in Fig. 3.1 for a molecule of iodine.

Show outer shell electrons only.



I_2
Br_2
Cl_2

Fig. 3.1

[2]

2 Potassium iodide is an ionic compound.

(d) Aqueous chlorine reacts with aqueous potassium iodide.

(i) Complete the symbol equation for this reaction.

$Cl_2 + 2KI \rightarrow I_2 + 2KCl$ [2]

F
Cl
Br
I

(ii) Choose from the list the name of this type of reaction.

Draw a circle around your chosen answer.

addition **combustion** **displacement** **neutralisation** [1]

(iii) State the colour of chlorine gas at room temperature and pressure.

Green ..... [1]

3 **(a)** Table 2.1 shows some properties of the halogens.

* MP & BP increases

Table 2.1

halogen	melting point in °C	boiling point in °C	density at room temperature and pressure in g/cm³
fluorine	–220	–188	0.0016
chlorine	–101	–35	0.0032
bromine		+59	3.1
iodine	+114	+184	

M.P inc

density increases

B.P 59 gas

Use the information in Table 2.1 to predict:

(i) the melting point of bromine

between -101 & 59 (excluding -101 and 59) ..... [1]

(ii) the density of iodine at room temperature and pressure

more than 3.1 ..... [1]

(iii) the physical state of chlorine at –10 °C. Give a reason for your answer.

MP → -101°C
BP → -35°C

physical state Gas

reason -10°C is above b.p

..... [2]

(b) The equation for the reaction of aqueous chlorine with aqueous potassium iodide is shown.

$$Cl_2 + 2KI \rightarrow I_2 + 2KCl$$

F
Cl
Br
I

(i) Choose the word which best describes this type of chemical reaction.
Draw a circle around your chosen answer.

addition **displacement** **neutralisation** **polymerisation** [1]

(ii) Explain why aqueous iodine does **not** react with aqueous potassium chloride.

chlorine is more reactive than iodine ..... [1]

4 **(a)** Table 7.1 shows some properties of some of the halogens.

Table 7.1

halogen	melting point / °C	boiling point / °C	colour
chlorine	−101	−35	yellow-green
bromine	−7		red-brown
iodine	+114	+184	grey-black
astatine	+302	+337	

Use the information in Table 7.1 to suggest:

- the colour of astatine ..... Black .....
- the boiling point of bromine ..... btw −7 & 184 (excluding −7 and 184) .....
- the state of iodine at 190 °C. ..... gas .....

[3]

(b) Aqueous chlorine reacts with aqueous potassium bromide as shown.

$$Cl_2 + 2KBr \rightarrow Br_2 + 2KCl$$

(i) Name the salt formed in this reaction. → Potassium chloride

..... [1]

(ii) Explain why aqueous bromine does **not** react with aqueous potassium chloride.

Bromine is less reactive than chlorine

..... [1]

(iii) Complete the dot-and-cross diagram in Fig. 7.1 of a molecule of chlorine.

Show outer shell electrons only

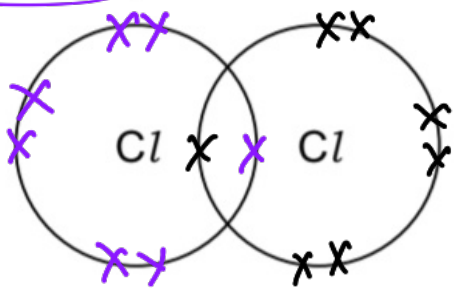


Fig. 7.1

[2]

[Total: 7]

✱ Transition Elements

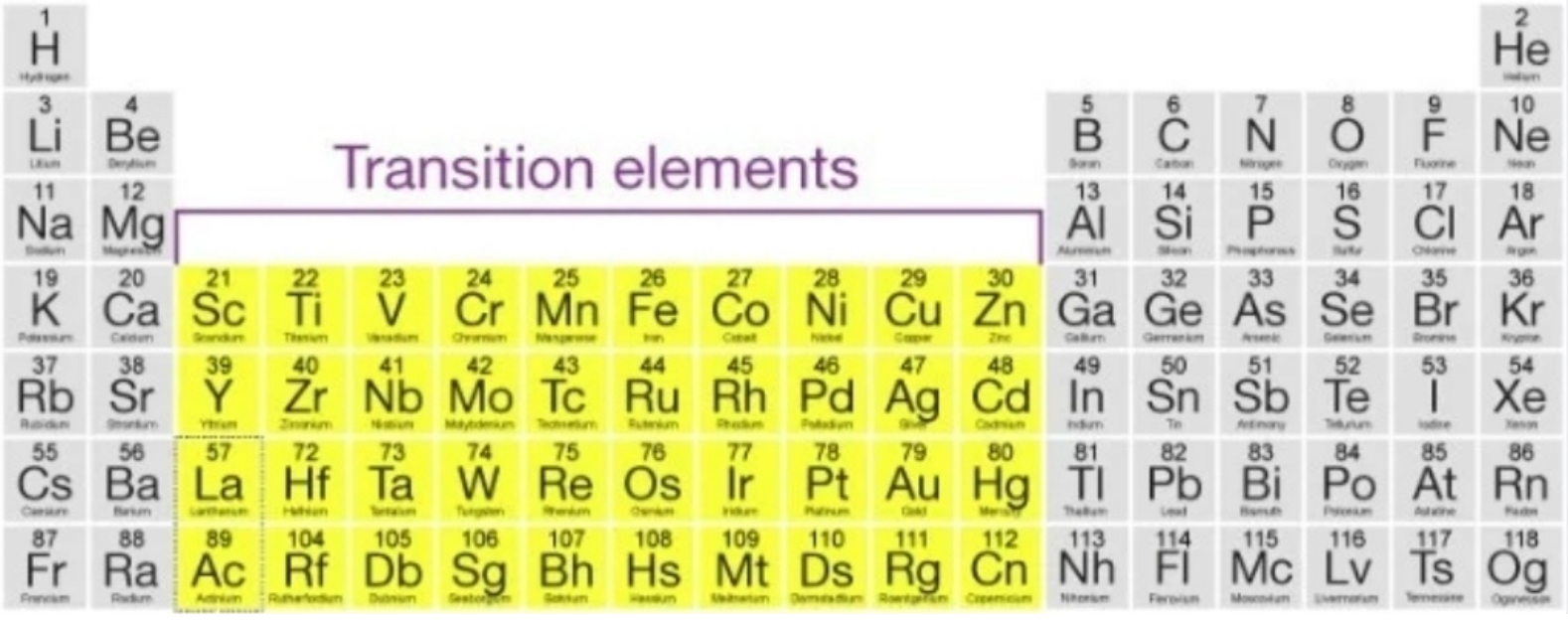
- They are found between group 2 and group 3
- Physical properties –
  1. They are lustrous
  2. They are hard and strong
  3. They conduct heat and electricity
  4. They have high M.P
     except mercury [liquid at room temp]

Few more properties –

1. They form ions with different charges
   Eg Fe^{+2} Fe^{+3}

2. They are useful as catalysts
   Eg Iron is used in Haber's process

3. They form coloured compound



Paper 3

Questions are applicable for both core and extended candidates

1 Samarium is a metal.

(a) Deduce the number of electrons and neutrons in the samarium atom shown.

$^{154}_{62}Sm$

p+n

p=e

number of electrons ..... 62 .....

number of neutrons ..... 154 − 62 = ...

[2]

(b) Samarium has properties that are similar to the properties of transition elements.

Choose **one** statement about samarium that is correct.

Tick (✓) **one** box.

- Compounds of samarium are colourless. ☐
- Samarium has a low melting point. ☐
- Samarium and its compounds do **not** act as catalysts. ☐
- Samarium has a high density. ☑

[1]

2 This question is about metals and compounds of metals.

(a) Table 4.1 shows some properties of five metals, **A**, **B**, **C**, **D** and **E**.

Table 4.1

metal	density in g/cm^3	melting point in °C	colour of metal chloride
A	5.90	30	white
B	5.96	1890	green
C	11.34	328	white
D	8.90	1455	yellow
E	1.53	39	white

State which **two** of these metals, **A**, **B**, **C**, **D** and **E**, are transition elements.

Give two reasons for your answer using only the information in Table 4.1.

metals ..... B ..... and ..... D .....

reason 1 ..... coloured compounds .....

reason 2 ..... High M.P .....

[3]

3 This question is about metals.

(a) Iron is a transition element. Potassium is an element in Group I of the Periodic Table.

State **two** differences in the physical properties of iron compared to potassium.

1 ..... Iron high density .....

2 ..... Iron high mp .....

[2]

5 (d) Cobalt is a transition element. Lithium is a Group I element.

Describe **two** ways in which the properties of cobalt differ from those of lithium.

1 .....

2 .....

[2]

6 This question is about metals.

(a) Nickel is a transition element. Sodium is an element in Group I of the Periodic Table. Nickel has a higher melting and boiling point than sodium.

Give two **other** ways in which the physical properties of nickel differ from the physical properties of sodium.

1 Nickel has higher density

2 Nickel forms colored compound

[2]

7 Copper is element 29 in the Periodic Table.

(c) Copper is a transition element.

Some physical and chemical properties of transition elements are shown.

physical properties:
- high density
- high strength

chemical properties:
- form coloured compounds
- have ions with variable oxidation numbers

(i) State one **other** physical property of transition elements.

High mp & bp ..... [1]

(ii) State one **other** chemical property of transition elements.

act as catalyst ..... [1]

8 Copper is a transition element. It has variable oxidation states.

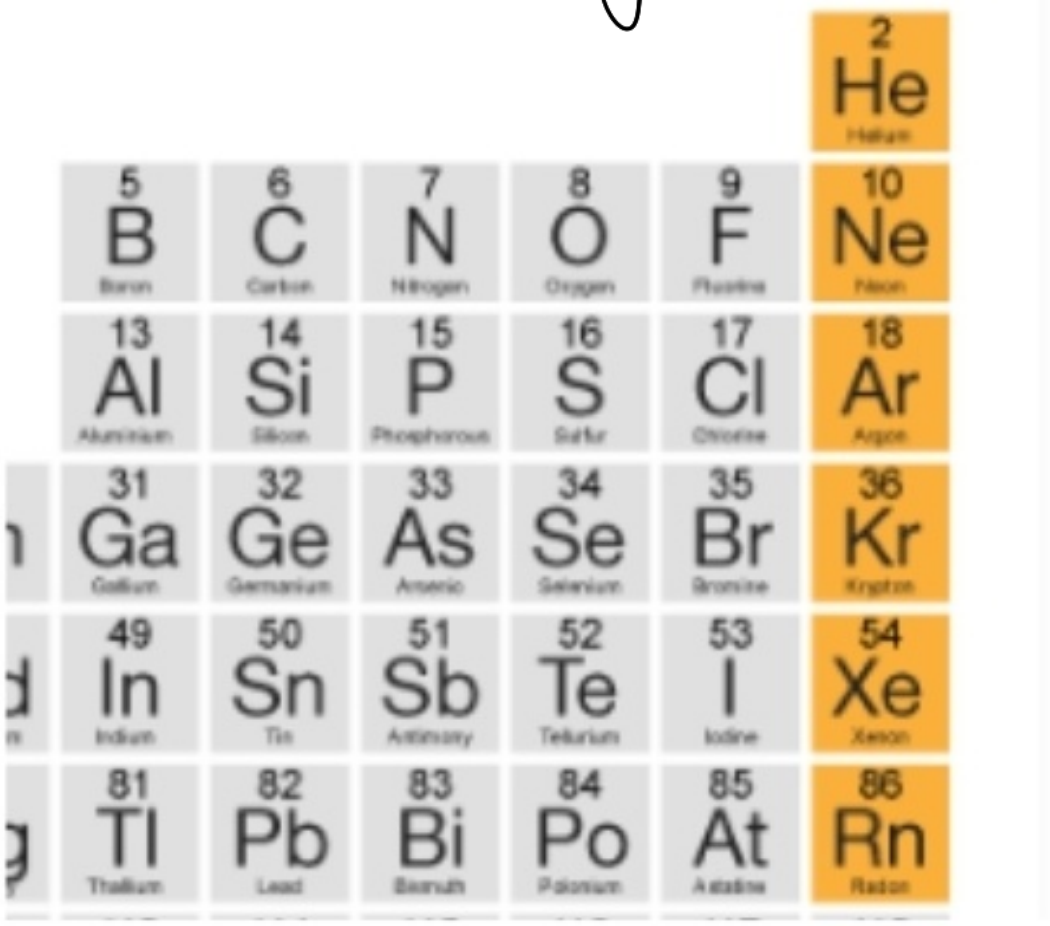
(a) State **two** other chemical properties of transition elements which make them different from Group I elements.

1 act as catalyst

2 variable valency / form ions with different charges

[2]

Noble gasses –



1 The electronic configurations of four elements, P, Q, R and S, are shown.

element	electronic configuration
P	2
Q	2,2
R	2,6
S	2,8

Which elements are unreactive monatomic gases?

A P and Q **B** P and S **C** Q and R **D** S only

Noble Gasses –

- The noble gasses are monoatomic
  made up of single atom
- They are colourless
- They are gaseous
- They have very low M.P/B.P
- ⭐ They are unreactive / inert

- They have full outer shells
- Electronic config of noble gas

He	2				
Ne	2	8			
Ar	2	8	8		
Kr	2	8	18	8	
Xe	2	8	18	18	8

- This means that noble gas no need to gain or lose electrons
- Electronic config extremely stable
- They do not easily form bond with other elements